

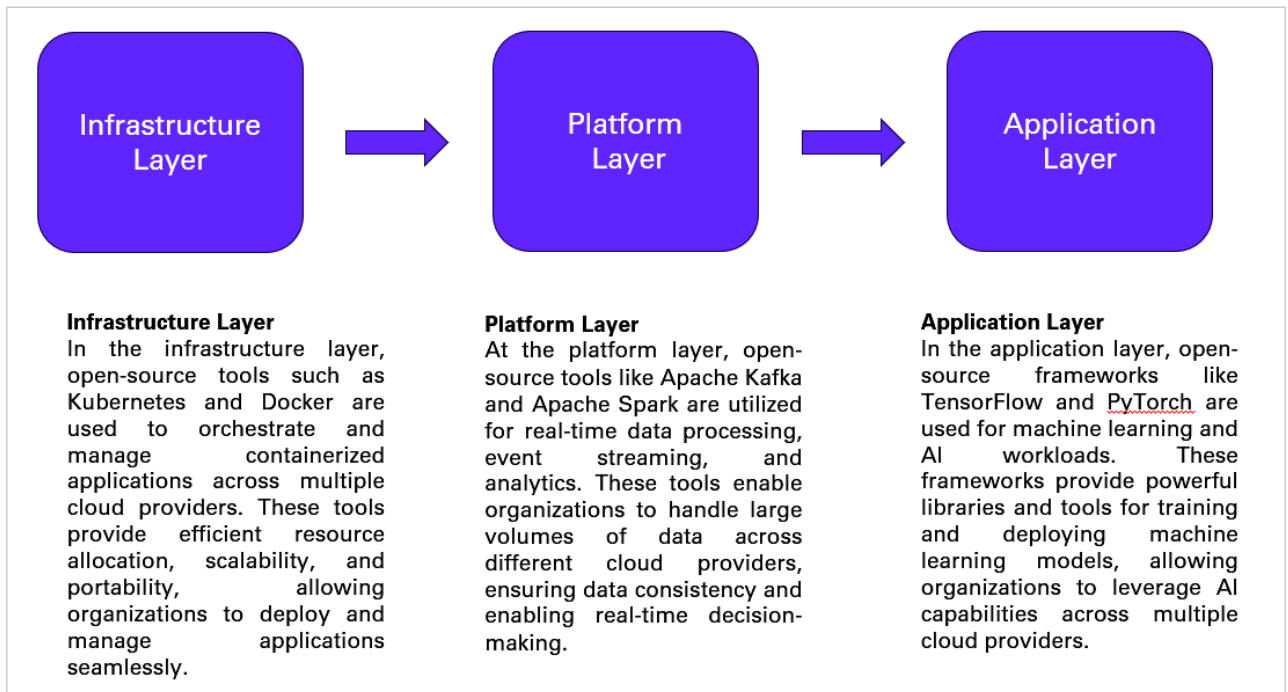


Figure 1: Role of open source tools in multi-cloud architecture

However, managing multi-cloud environments introduces new challenges.

- **Complexity:** Coordinating configurations, deployments, and operations across diverse platforms
- **Vendor lock-in:** Risk of reliance on a provider, limiting agility and cost advantages
- **Security:** Ensuring consistent security policies and procedures across different cloud environments
- **Governance:** Maintaining visibility, control, and compliance across cloud deployments

Open source: The key to unlocking multi-cloud agility

Open source tools play a crucial role in each layer of the multi-cloud architecture, providing flexibility, scalability, and cost-effectiveness. These tools enable organisations to leverage the benefits of multi-cloud environments while avoiding vendor lock-in and promoting interoperability.

Open source tools empower organisations to address these

challenges by offering:

- **Cost-effectiveness:** Freely available or with lower licensing costs, reducing infrastructure expenses
- **Customisation:** Adaptable to specific needs and workflows, promoting flexibility and innovation
- **Vendor neutrality:** Avoid lock-in, enabling switching between providers for optimal cost and performance
- **Community-driven development:** Continuous improvement, bug fixes, and security updates through active communities

Essential open source tools for your multi-cloud AI toolkit

- **Infrastructure as Code (IaC):** Terraform, Ansible, and Pulumi codify deployments for consistency, repeatability, and automation
- **Container orchestration:** Kubernetes manages containerised workloads across clusters, simplifying AI model deployments
- **Monitoring and logging:** Prometheus, Grafana, and

Elasticsearch provide unified, real-time insights into multi-cloud resources and AI models

- **Cost management:** CloudHealth and OpenStack CloudKitty track, analyse, and optimise cloud spending across providers
- **Security and compliance:** OpenFaaS and Knative provide serverless functions for secure and scalable AI deployments
- **CI/CD pipelines:** Jenkins and GitLab automate AI development and deployment life cycles across clouds

Best practices for open source multi-cloud AI success

- **Adopt a cloud-native approach:** Design cloud-agnostic AI applications for portability and flexibility
- **Standardise on open APIs:** Facilitate communication and integration between AI models and services across clouds
- **Choose tools based on maturity and adoption:** Select well-established,